

Muhammad Hamza

Data Analyst & Automation Specialist

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SUMMARY

Data Analyst with 7+ years progressing from SEO and e-commerce into full-time analytics and automation. Built 15+ production Power BI dashboards and 8 end-to-end automation pipelines across 6 industries, spanning web scraping, AI-driven sentiment analysis, speech-to-text monitoring, and business intelligence. Specialized in turning manual reporting workflows into hands-off systems using Apache Airflow, Python, Selenium, Playwright, and Apps Script.

SKILLS

Power BI & Data Modeling: Power BI, Power Query (M), Data Modeling (star schema, fact & dimension tables, relationships, cardinality), Advanced DAX (calculated measures & columns, time intelligence), Drill-through, Interactive Filtering

BI & Visualization: Tableau, Looker Studio, Qlik Sense, Excel

Databases & SQL: MySQL, MS Access

Programming: Python, SQL, JavaScript (Google Apps Script), HTML, CSS, Bootstrap

Data & Scraping: Selenium, Playwright, BeautifulSoup, Pandas, DeepSeek API, OpenAI Whisper

Automation & AI Tools: Apache Airflow, Apps Script, Tkinter, MCP Servers, N8N, Zapier, Landbot

Research & Monitoring: SPSS, IPSOS Synthesio, Brand24

E-Commerce: Helium 10, Keepa, Jungle Scout, E-Rank, Marmalead, Mailchimp

EXPERIENCE

Data Analyst

Apr 2023 – Jun 2026

Press Information Department

- Designed and developed interactive dashboards in Power BI using multiple datasets to visualize performance metrics and surface actionable insights.
- Built the underlying data models in Power Query — cleaning, shaping, and merging multiple source datasets — and designed star-schema models with fact and dimension tables, defined relationships, and cardinality to keep reports scalable.
- Authored advanced DAX measures and calculated columns — time intelligence, weighted averages, and ratio calculations — to power KPIs, trend analysis, and segmentation across the dashboards.
- Integrated Power BI with Microsoft Access, Excel, and SQL Server to establish seamless data connections and reporting pipelines.
- Automated the complete daily reporting process in Python, significantly reducing manual workload and improving operational efficiency.
- Built an end-to-end Apache Airflow pipeline that scrapes daily coverage, runs each item through the DeepSeek API for sentiment analysis, and generates a daily media monitoring report in Word format.
- Developed multiple Python scrapers (Selenium, Playwright, BeautifulSoup) to extract data from 29+ newspapers — collected clean text into JSON/CSV ready for analysis.
- Built a Tkinter + Selenium tool that automates Brand24 monitoring end-to-end — pastes keywords, applies region/language/date filters, downloads the PDF report, renames it, posts it to the relevant WhatsApp group, and archives the file to prevent duplication.
- Created an urgent-case alert pipeline using Google Apps Script and Python — triggered by a Google Form, it generates a full PDF report, emails stakeholders, drops the file in Drive, and a Python watcher (scheduled via .bat every 5 minutes) ships it to the WhatsApp response group.
- Implemented a multi-language speech-to-text transcription system using OpenAI Whisper.
- Built a YouTube sentiment analysis solution using a Model Context Protocol (MCP) server — automated evaluation of public sentiment across video content and audience comments.
- Applied data extraction and analytical techniques to identify, analyze, and interpret significant trends across diverse data sets.
- Ensured data integrity and adherence to data security and compliance standards across all analysis and reporting tasks.

E-Commerce Specialist (Team Lead)

Aug 2022 – Dec 2022

Vapes Direct

- Built Power BI dashboards for sales and performance metrics, driving data-informed decisions across the team.

- Led WordPress e-commerce operations — product listings, inventory management, acquisition tracking, conversion optimization, and A/B testing.
- Managed the company ERP system for stock-level tracking, inventory management, and purchase orders — keeping stock records accurate and aligned with live e-commerce operations.
- Maintained Standard Operating Procedures (SOPs) and dispatching plan for the e-commerce department.

E-Commerce Officer

Apr 2020 – Jun 2022

Izoc Solutions

- Built monthly and annual sales reports and interactive dashboards in Excel and Power BI.
- Managed PPC campaigns across Amazon, Walmart, eBay, and Etsy — optimizing ad performance and ROI across four major marketplaces.
- Conducted keyword research and listing optimization to improve product ranking, click-through rates, and conversions.
- Performed product hunting (wholesale model) and pricing analysis to identify profitable, sustainable products with competitive positioning.
- Managed account health, case resolution, and platform policy compliance across all marketplaces.

SEO Specialist

Mar 2017 – Feb 2020

IT Supersonics

- On-page SEO optimization (HTML, CSS) to improve website rankings, structure, and search visibility.
- Keyword research and competitor analysis to guide SEO strategy and content optimization.

PROJECTS

HR Analytics — Employee Attrition Dashboard. Power BI dashboard across a 1,470-employee workforce — DAX-calculated risk multipliers surface attrition drivers (OverTime 3× risk, low income 5× risk, Sales Reps 39.8% attrition). Includes drill-through from any job role to employee-level detail, multi-dimensional segmentation (department, age group, marital status, income), and interactive filters.

Amazon Product Analytics Dashboard. Power BI dashboard analyzing 1,351 products with 23.8M total reviews across 4 main categories — surfaced 10–20K price tier driving 28% of GMV and 20K+ premium tier pulling 25.7%. Built with DAX measures for weighted ratings (4.16 vs 4.09 simple avg) and GMV proxy calculations, with drill-through to product detail view.

Sales & Supply Performance Dashboard. B2B Power BI dashboard integrating sales and purchasing databases — tracks YTD revenue, perfect order rate, on-time delivery, top customers by revenue, and 90-day demand-supply gap (5,821-unit shortfall flagged across SKUs) with interactive filters.

Esports Career Performance Analytics (Tekken 8). Power BI dashboard analyzing 273 matches across 32 tournaments worldwide — opponent matchup analysis, character-pick performance, tournament-by-tournament progression, and DAX-calculated win streak metrics. Time-series and segmentation techniques applied to esports data.

Fatwa Archive Scraper. Site-specific Python scrapers across multi-language fatwa archives — each handled the site's own structure and language quirks, producing ~300K clean JSON/CSV rows ready for analysis. Tech: Python, BeautifulSoup, Playwright.

Pulmonary TB Knowledge-Attitudes-Practices (KAP) Study. Pro bono quantitative analysis of a 250-respondent survey in SPSS — frequency distributions and demographic cross-tabs (gender, age, education, residence) supporting a medical student's thesis on care-seeking delays and TB awareness gaps.

EDUCATION

Bachelor in Computer Sciences

Feb 2014 – Feb 2018

The University of Lahore

CERTIFICATIONS

- Internet / Social Media Monitoring and Analysis · Aug 2023
- Sponsored Ads Certification — Amazon · Dec 2022
- Certified Ethical Hacking · Dec 2018 – Feb 2019